Lab 2 — Building Version 1 and Publishing to GitHub

After Module 2 — GitHub Copilot Introduction

Open the TodoApp folder from Lab 1 in VS Code. Confirm docs/PRD.md and docs/TRD.md are present, and that Python 3.12, Node 20, and Git are installed.

Step 1 — Switch to Agent mode

In Copilot Chat, set the mode picker to **Agent** and the model picker to Claude Sonnet 5.
Agent mode can create files, run terminal commands, and correct its own errors. Watch the chat panel as it works — you will be asked to approve terminal commands.

Step 2 — Build the application

Type this prompt and press Enter.

```
Build Version 1 of the Todo application exactly as specified in
#file:docs/TRD.md.

Follow the TRD precisely:
 - The folder structure in section 2.
 - The data model in section 3.
 - The API endpoints in section 4, including the status codes.
 - The three backend layers in section 5 - routers, services,
   repository. The repository is the only file that stores data.
 - The frontend components in section 6.

Additional constraints:
 - Backend: Python 3.12, FastAPI, Pydantic. Create a requirements.txt.
 - Store todos in a Python list inside the repository layer only.
 - Frontend: React created with Vite, JavaScript (not TypeScript),
   functional components, useState.
 - Enable CORS on the backend so the React dev server can call it.
 - Write the code like a junior developer would - simple and readable.

Do NOT include:
 - Any database or ORM
 - A Category field
 - Authentication or authorization
 - Exception handling beyond what FastAPI provides by default
 - Docker, tests, or CI configuration

When the code is written, create a README.md at the root with the exact
commands to run the backend and the frontend.
```

Approve the terminal commands when prompted. This will take several minutes.

Step 3 — Run and test the application

Open two terminals in VS Code (Ctrl + Shift + backtick).

Terminal 1 — backend:

```
cd backend
pip install -r requirements.txt
uvicorn main:app --reload
```

Terminal 2 — frontend:

```
cd frontend
```

```
npm install
npm run dev
```

Open the URL shown by the frontend terminal, usually http://localhost:5173. Also open http://localhost:8000/docs to see the FastAPI interactive API documentation.
Add three todos. Mark one complete. Delete one. Refresh the page — the todos are still there.
Now stop the backend terminal with Ctrl + C and start it again. Refresh the browser. The todos are gone. This is Version 1 behaving exactly as the PRD specified. Version 2 will fix it.

Step 4 — Fix anything broken

If the application does not run, do not fix it by hand. Select the error in the terminal, then in Copilot Chat:

```
The application fails to start with this error:

#terminalSelection

Read the relevant files and fix the cause. Do not change the folder
structure defined in #file:docs/TRD.md.
```

Step 5 — Create the .gitignore

In Copilot Chat:

```
Create a .gitignore at the repository root suitable for a project with
a Python FastAPI backend and a React Vite frontend in the same
repository. Include Python virtual environments, __pycache__,
node_modules, build output, and editor folders.
```

Open the generated .gitignore and read it. Confirm node_modules/ and __pycache__/ are listed.

Step 6 — Initialise the repository

In the terminal, from the TodoApp root:

```
git init
git add .
git status
```

Read the output of git status carefully. If node_modules or __pycache__ appear in the list, your .gitignore is wrong or was added too late. Fix it before continuing.

```
git commit -m "Version 1: Todo app with in-memory storage, built from TRD"
```

Step 7 — Create the GitHub repository and push

- On github.com click + in the top right, then New repository.
- Name it TodoApp. Set it to Private. Do not add a README, .gitignore, or licence.
- Click Create repository.

Copy the commands under "push an existing repository" and run them:

```
git remote add origin https://github.com/<your-username>/TodoApp.git
git branch -M main
git push -u origin main
```

Refresh the GitHub page. Confirm docs/PRD.md, docs/TRD.md, backend/, and frontend/ are all present.

What you should have at the end of this lab

```
TodoApp/
  .gitignore
  README.md
```

```
docs/PRD.md
docs/TRD.md
backend/
frontend/
```

Running locally and pushed to GitHub.